

# Test 4

May 7, 2024

## Problem 1

A barn has dimensions  $30 \times 50$  ft. A cow is tethered to a corner of the barn by a rope 60 ft long. What is the locus of points where the cow may graze?

## Problem 2

Let  $A$ ,  $B$ , and  $C$  be 3 noncolinear points. What is the locus of points equidistant from them?

**Problem 3**

A cylinder has a radius of 8 in and a height of 12 in. Find a) its surface area and b) its volume.

**Problem 4**

A sphere has a radius of 5 ft. Find a) its surface area and b) its volume.

### Problem 5

A snowman is composed of three spherical snowballs and a cone carrot for the nose. If each of the snowballs has a radius of 9 in, and the carrot is 10 in long with a base radius of 1 in, what is a) the volume of the snowman and b) the surface area of the snowman?

### Problem 6

Calculate the distance between the following sets of points:

a)  $(0, -3)$  and  $(0, 9)$

b)  $(0, 0)$  and  $(3, 4)$

c)  $(-2, -9)$  and  $(9, 3)$

d)  $(10, -3)$  and  $(4, 5)$

**Problem 7**

Calculate the midpoint between the following sets of points:

a)  $(0, -3)$  and  $(0, 9)$

b)  $(0, 0)$  and  $(3, 4)$

c)  $(-2, -9)$  and  $(9, 3)$

d)  $(10, -3)$  and  $(4, 5)$

**Problem 8**

Solve the right triangle with a leg length of 6, and whose hypotenuse has a length 10.

### Problem 9

Calculate the following:

a)  $\sin\left(\frac{3\pi}{2}\right)$

b)  $\cos\left(\frac{\pi}{6}\right)$

c)  $\sec\left(\frac{11\pi}{6}\right)$

d)  $\tan\left(\frac{3\pi}{4}\right)$

### Extra Credit

Prove that the angle sum of any triangle is a straight angle.